CS & IT ENGINEERING

Algorithms

Graph Algorithms

Lecture No.- 01



By- Aditya sir

Recap of Previous Lecture



Topic Travelling salesman

Topic PYQs

Topics to be Covered



Topic Graph Traversals

ABOUT ME

Hello, I'm Aditya.

1. Represented college as the first Google DSC Ambassador.
2. The only student from the batch to secure an internship at Amazon.
3. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis8. Completed my Masters with an overall GPA of 9.36/108. Joined Dream11 as a Data Scientist.9. Have mentored around 10k working professions in field of Data Science and Analytics10. Have been mentoring GATE aspirants to secure a great rank in limited time.11. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.



By- Aditya sir



Topic : Graph Traversals

Graph Traversals:

Graph traversal is a process of visiting each and every node of the tree/graph in some specific order and processing the information exactly once.



Topic : Graph Traversals

Graph Traversals:

(1) Depth First Search (DFS)/ Traversal (DFT)

(a) Undirected graph

I. Connected

II. Disconnected

(b) Direct graph

I. Directed acyclic graph (DAG)

II. Topological Sorting /ordering



Topic : Graph Traversals

Graph Traversals:

(2) Breadth First Search (BFS)/Traversal (BFT)

(a) FIFO BFS (Default / standard Approach)

(b) LIFO BFS (Bonus)



Topic : Graph Traversals

DFS ⟶ Stack ⟶ LIFO

BFS ⟶ Queue ⟶ FIFO



Topic : Graph Traversals

Applications of DFS and BFS:

- → Topological Sorting
- → Connected Components (CC)
- → Strongly Connected Components (SCC)
- → Articulation point / cut vertex
- → Biconnected Components (BCC)
- → Properties

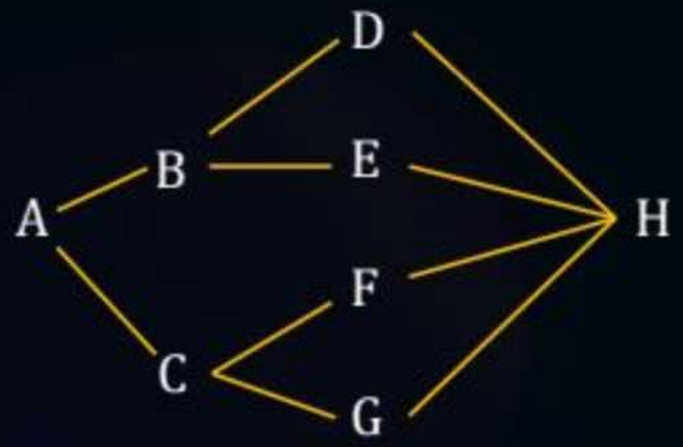


Topic : Graph Traversals

(1) Breadth First Traversal (BFT)

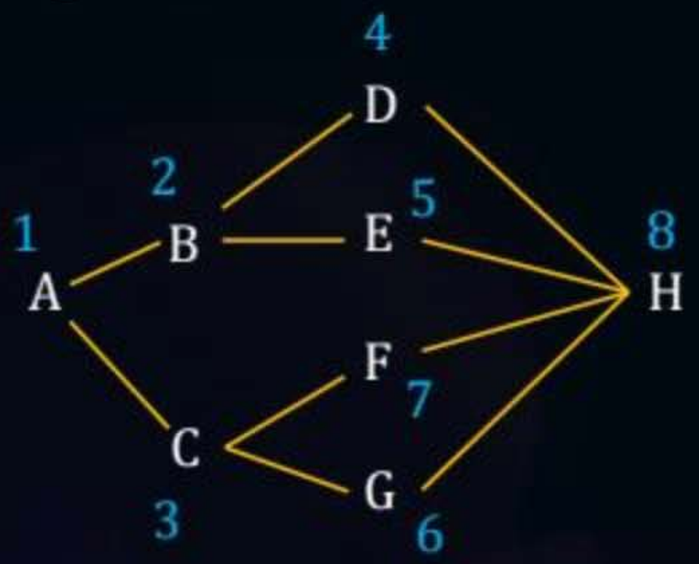
(a) FIFO BFS/BFT (default)

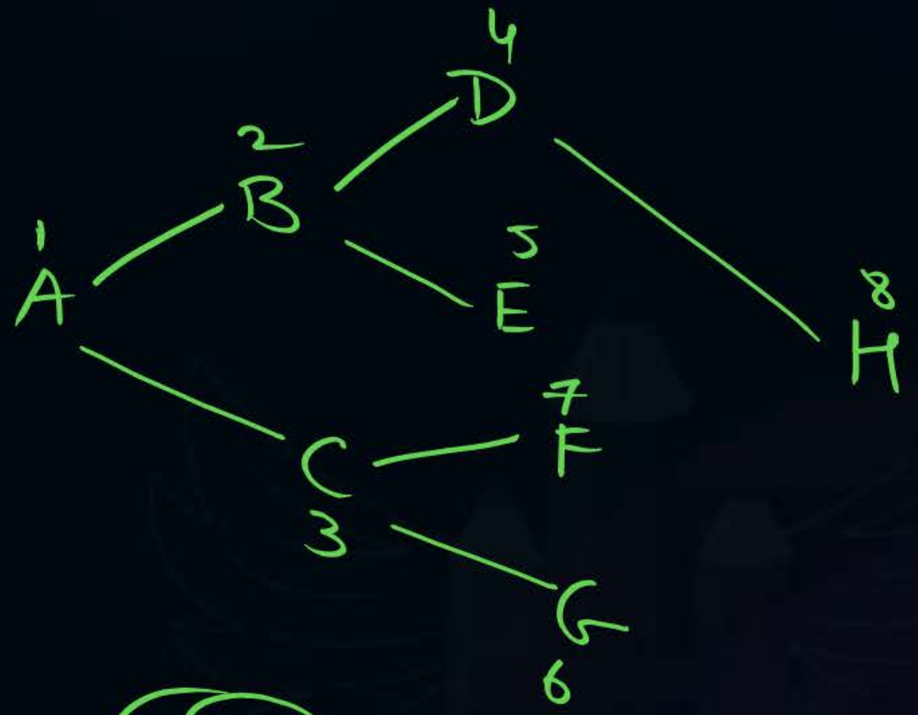
e.g. Given



Start at A

BFS Spanning Tree





Queue

Node	B	C	D	E	G	F	H
parent	A	A	B	B	C	C	D

output: ABCDEGFH

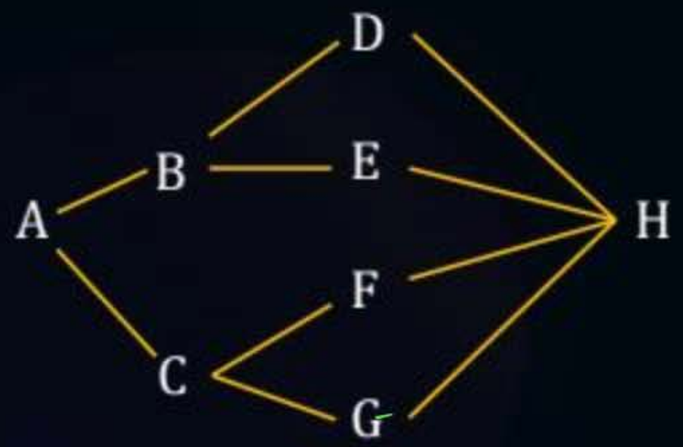


Topic : Graph Traversals

Observations: (V. V. Imp)

Assume: every edge cost = 1

Gate 25



1

A – B – D – 2

	A ⟶ D	
1.	A – B – D	1 + 1 = 2
2.	A – B – E – H – D	1 + 1 + 1 + 1 = 4
	and many more	



Topic : Graph Traversals

Property:

In a uni-weighted graph (where every edge has equal weight) / un-weighted

BFS solves the problem of (SSSP) single source shortest paths from starting vertex to all other vertex.



Topic : Graph Traversals

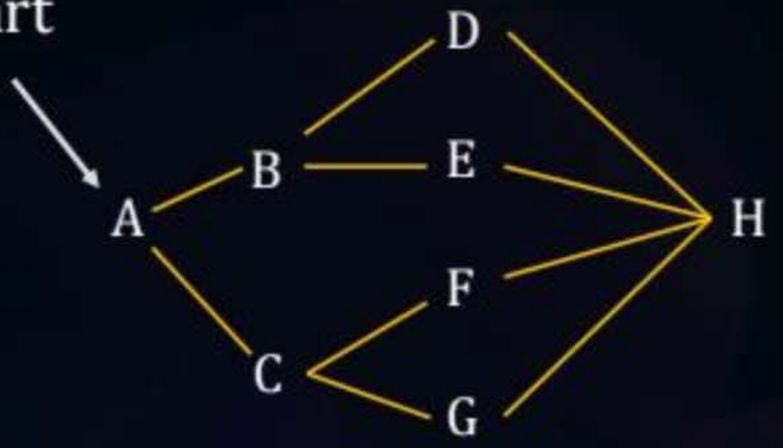
(Q) Is it same as DFS?
→ No

Bonus:

LIFO BFS (Special case)

There we are a Queue that follows LIFO order (Last in first out)

Start





Queue

Node	B	C	E	C	H	B	E
parent	A	A	C	C	G	H	H

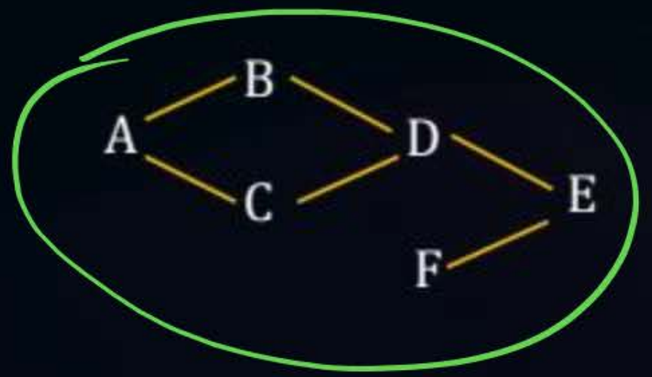
output: ACGHEDFB



Topic : Graph Traversals

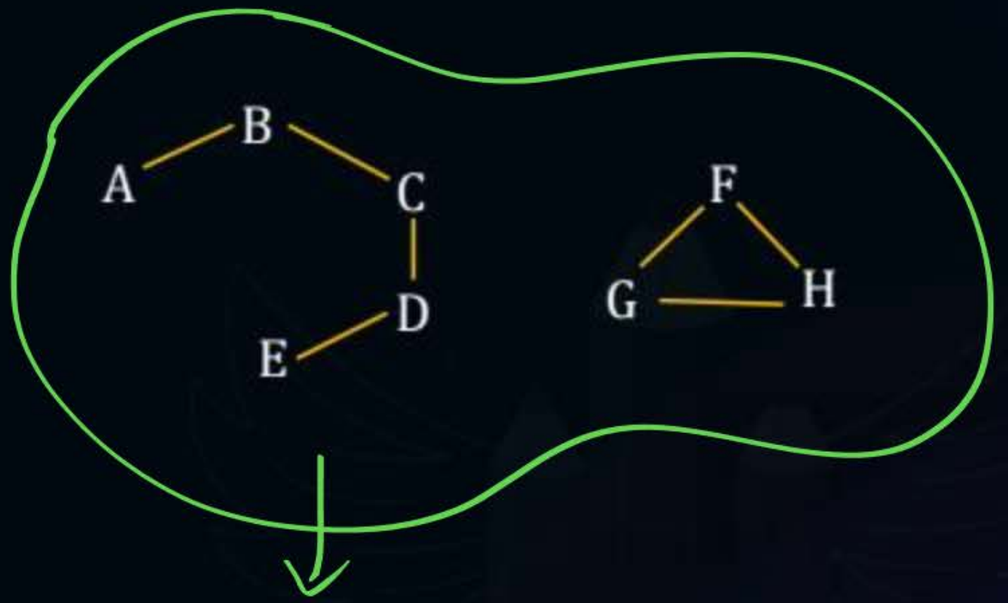
Depth First Traversals (DFT/DFS)

1. Connected graph



1 Component

(2) Disconnected graph



2 Components



Topic : Graph Traversals

Depth First Traversal (DFT)

(a) DFS/DFT on an undirected Graph;

I. E-node (Exploring node)

→ Node which is currently getting explored.

II. Live Node.

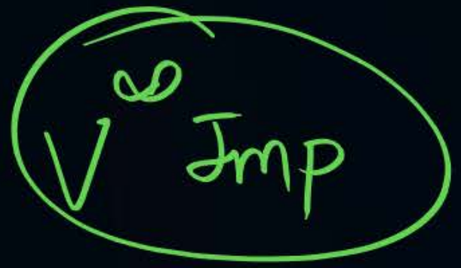
→ Node that is NOT yet fully explored. (live nodes are present in stack)

III. Dead Node.

→ Node that has already been fully explored (already popped out from stack).



Topic : Graph Traversals



During Traversal, the different types of timing values that a node is associated with.

(1) Discovery time: d(n)

→ It is the time at which the node has got discovered/visited for the very first time.

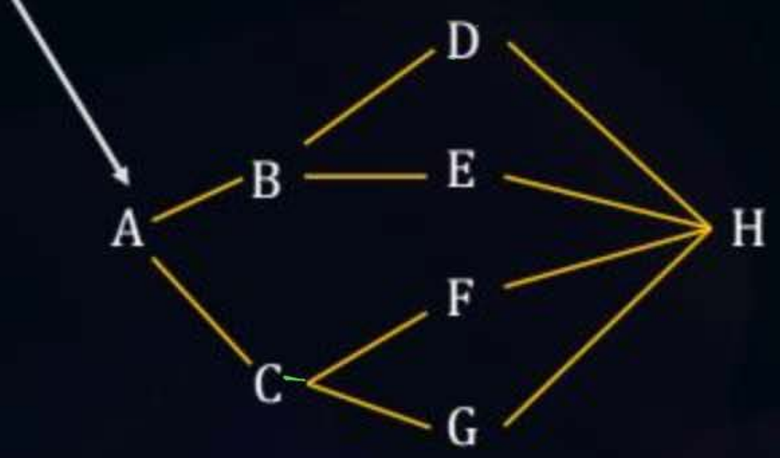
(2) Finishing time: t(n)

→ It is the time at which the node becomes a dead node.

Topic : Graph Traversals

DFS in undirected connected graph:

Start



ABDHEFCG



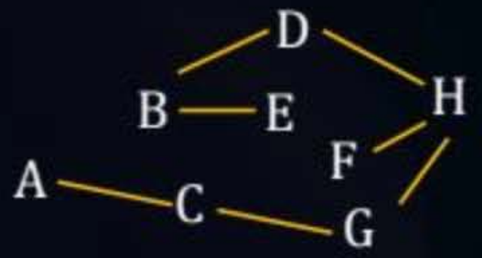


Stack

output: ABDHEFCG

Stack empty ⟶ **End of DFS**

Multiple BFS, DFS / ~~req~~ are possible



ACGHDBEF ⟶ also valid DFS starting from A.

$n = 4$ Complete graph

How many BFS?

$n!$

$4 \times 3 \times 2 \times 1 \Rightarrow 4!$ ✓

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#Q. Which of the following represent valid DFS traversal on graph G?

BFS

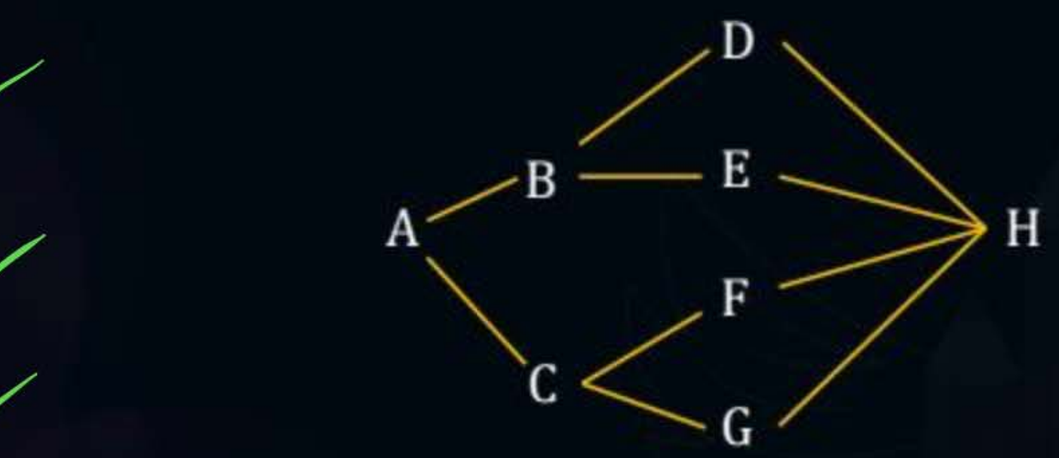
A. A B F E D H G C

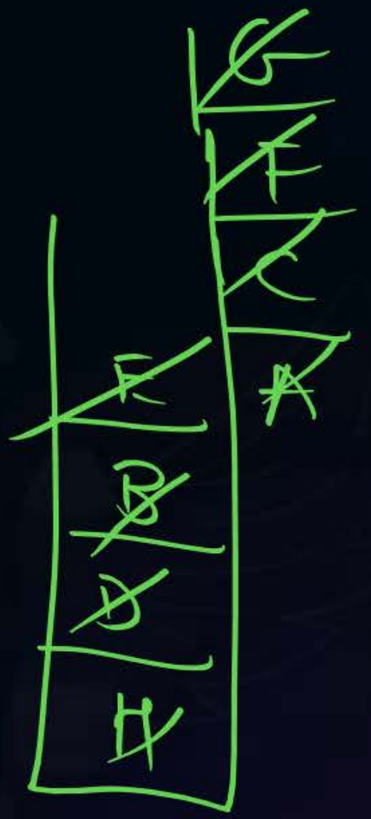
B. A B E H D F C G

C. D H G C F A B E

D. H E B D A C G F

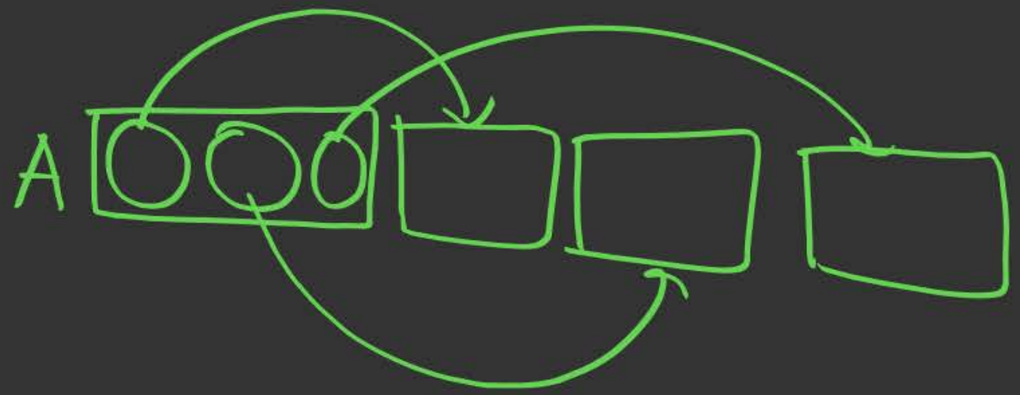
E. H D B E F C G A



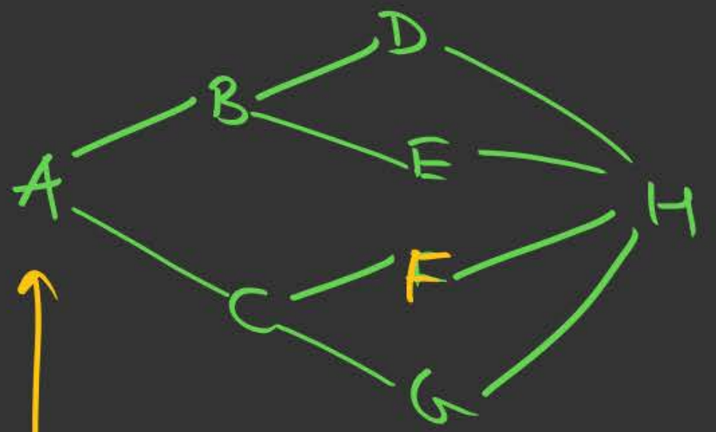


Bonus

BFS



Discovery & Finishing time :



Starting node (A)

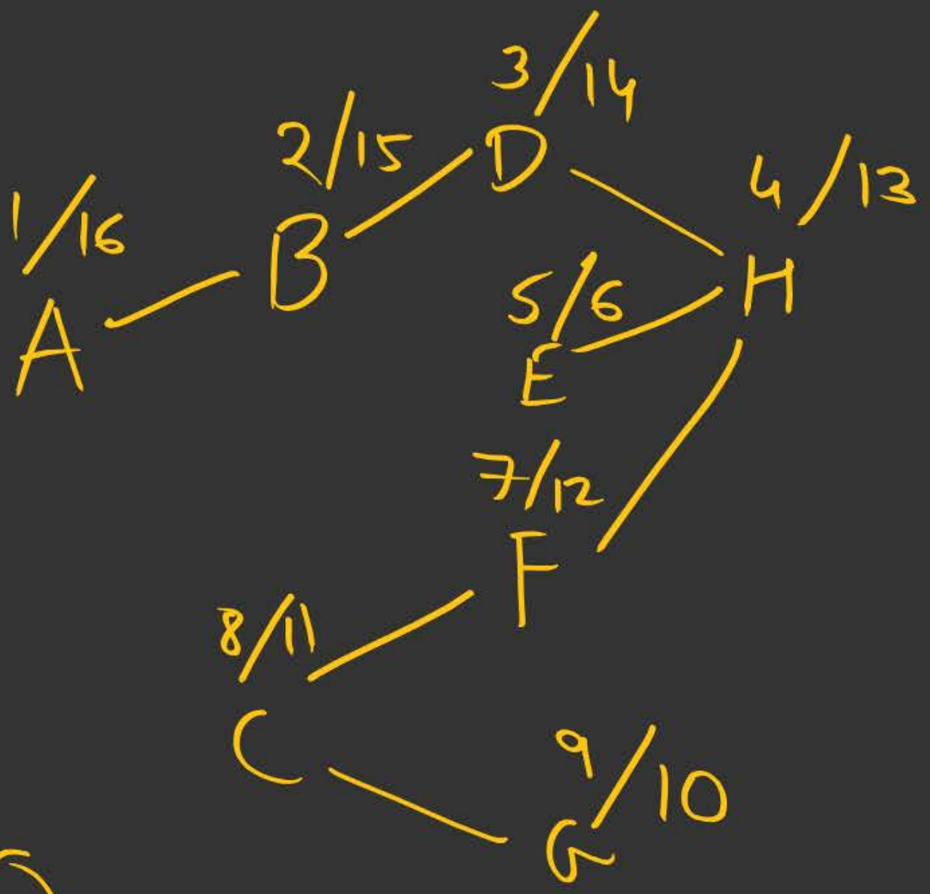
d/f

Node

1 →

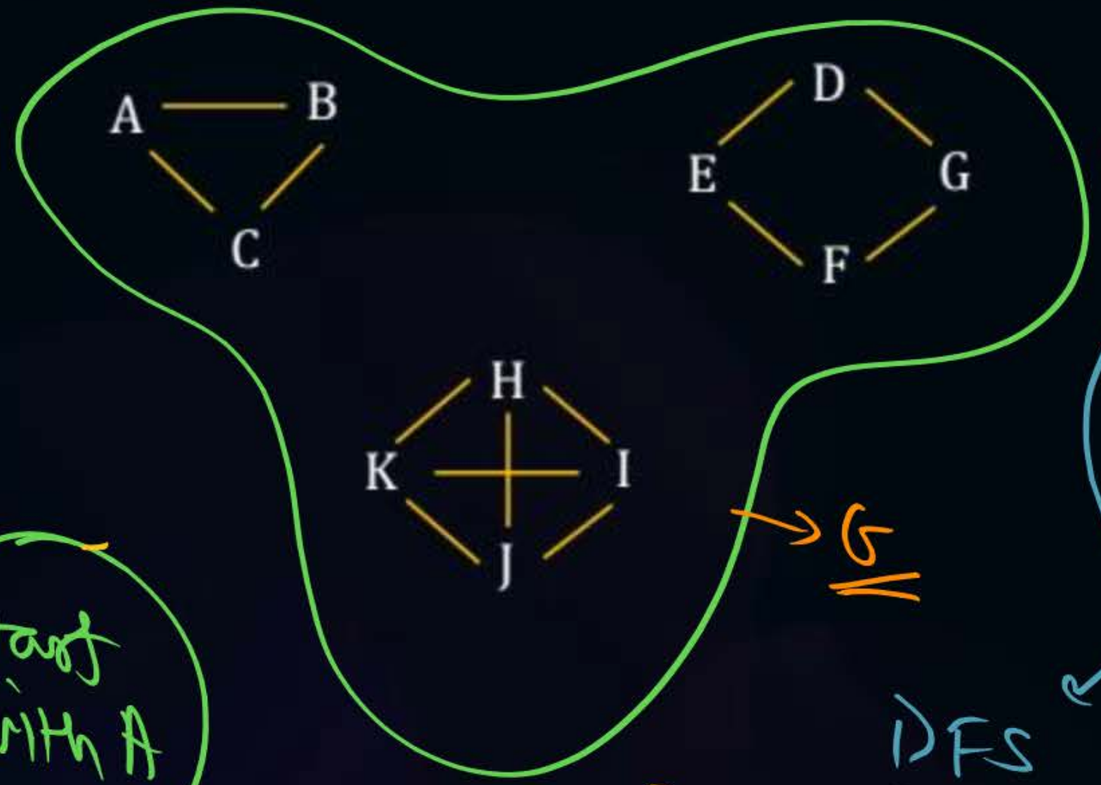
n = 8

2 × 8 = 16



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#Q. DFS / DFT on a undirected Disconnected graph.



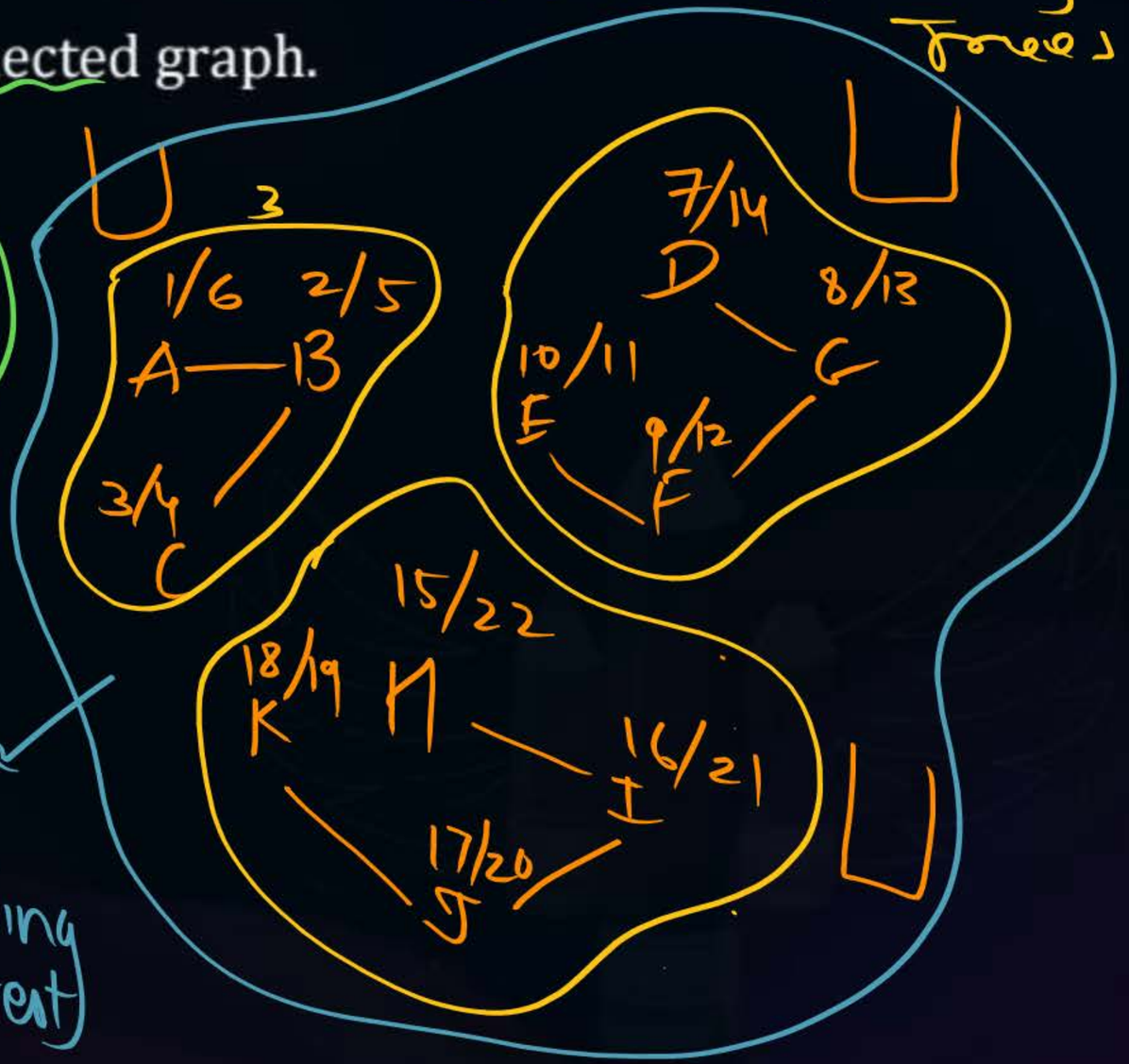
G

Start with A

3 CC

DFS Spanning Forest

3 DFS Spanning Forests





Topic : Graph Traversals

V.V.Imp Application of discovery & Finishing time

#Q. You are given a graph (G) with 5 vertices {A, B, C, D, E}
The Discovery and Finishing times for each node/vertex after carrying out DFS are as follows.

10

(V^2 Imp)

1



Topic : Graph Traversals

#Q. You are given a graph (G) with 5 vertices {A, B, C, D, E}
The Discovery and Finishing times for each node/vertex after carrying out DFS are as follows.

1. How many connected components are in G?



Topic : Graph Traversals

#Q. You are given a graph (G) with 5 vertices {A, B, C, D, E}
The Discovery and Finishing times for each node/vertex after carrying out DFS are as follows.

2. What are the vertices in each of those connected components ?
(for each option)

(d, f)	A	B	C	D	E
1.	(1, 2)	(3, 5)	(7, 8)	(11, 12)	(18, 20)

→ Disconnected
⑤ A, B, C, D, E

2. (1, 15) → (3, 13) → (4, 12) → (5, 11) → (6, 9)

→ Connected ①
{A, B, C, D, E}

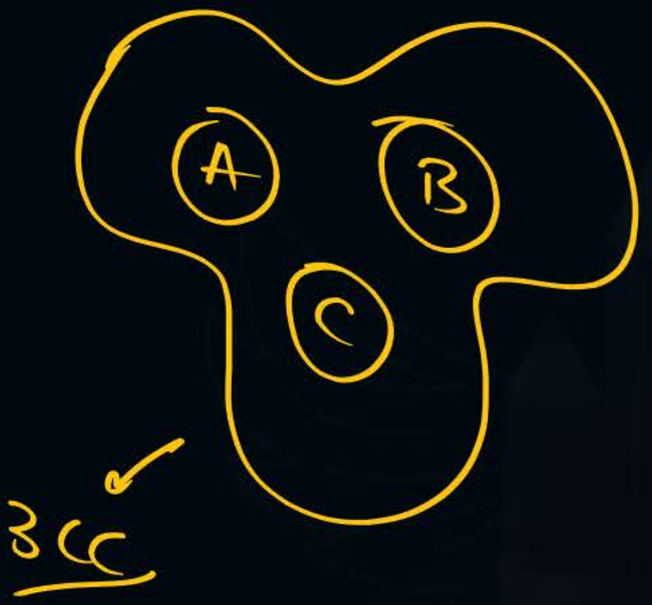
★3. (1, 14) → (2, 7) (16, 18) (19, 25) → (21, 22)

→ Disconnected ③
{A, B}, {C}, {D, E}

4. (1, 16) → (3, 11) → (5, 10) (20, 25) → (22, 23)

→ Disconnected ②
{A, B, C} {D, E}

1 single node is also a connected component.



NOTE: In D/F times, its only mandatory to have time in increasing order

5 10 12 18



THANK - YOU